

# An Image Combination Segmentation Method Based on Clustering Analysis and Edge Detection

Guocheng Liu\*

College of Information Engineering, Guangzhou Railway Polytechnic, Guangzhou, China  
c3c365@foxmail.com

## ABSTRACT

Considering that it is difficult to completely segment the spider mite image on the leaves of field crops from the leaf background, a combination segmentation method combining K-means clustering algorithm and Canny edge detection algorithm is proposed. This method first uses the K-means clustering algorithm to filter out most of the leaf background, then extracts the edge closed contour of the spider mite based on Canny edge detection, and implements the binarization segmentation of the spider mite image by algorithms such as seed filling and morphological opening operations. Experiments show that this method can achieve complete segmentation of spider mites images on leaves, which provides a new technique and method for spider mite pest analysis and insect number counting.

## CCS CONCEPTS

• **Computing methodologies** → Artificial intelligence; Computer vision.

## KEYWORDS

image segmentation, edge detection, spider mite image, K-means algorithm

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## 1 INTRODUCTION

Spider mite (commonly known as red spider) is an important economic pest mite that damages many important economic crops such as citrus, apple, grape, lychee, peach, lemon, mulberry, cotton, flowers, etc. [1-4]. At present, the collection and monitoring of field spider mite pest information mainly depends on manual counting [5]. Although this method is simple and easy to implement, it requires operators to have certain professional basic knowledge, and the use of manual visual statistical data is often prone to big error due to difference in operator's subjective judgment. In recent

years, with the continuous development of image acquisition equipment and computer image analysis technology, image recognition analysis enjoys the advantages of high accuracy, small labor and visualization, which has become a hot spot in the current insect recognition research [6]. For example, Qiu Daoyin et al. used the BP neural network to identify and analyze red spider image of jujube tree [7]; Wu Canghai et al. used Matlab multiple edge detection algorithms to comparatively analyze the citrus spider mite image [8]; Zhang Hua et al. used two-dimensional LWT wavelet theory to analyze and recognize the image of field spider mites [9]; Wang Jian et al. proposed a method to identify the damaged area from the hyperspectral image of the leaves damaged by red spiders [10]. These studies indicate that computer image processing and analysis technology has important application value and research significance in the field of spider mite control.

The first problem to be solved by insect image analysis is image segmentation [11]. Based on this, we study field spider mite image segmentation technology, and propose a combination segmentation method for spider mite image based on the combination of K-means clustering analysis and Canny edge detection. This method acquires the leaf image of spider mites pests through the mobile terminal image acquisition device and sends it to the computer image processing and analysis system. Afterwards, the image processing and analysis system completes the segmentation and extraction of spider mites image. The method provides a new technology and new method for subsequent spider mite identification analysis and insect number calculation.

## 2 DESIGN OF IMAGE ACQUISITION AND ANALYSIS SYSTEM FOR FIELD SPIDER MITES

Field image acquisition equipment needs to be portable, fast and mobile with built-in power. The image acquisition and analysis system adopts the architecture design of "mobile terminal + background server", as shown in Figure 1

The system uses mobile terminals to collect field pest images and perform image analysis. The collected images and analysis results can be transmitted to a remote background server via a wireless mobile network. Where, the mobile terminal serves as the main equipment for collecting field spider mite images. It consists of five modules: image collector, central processing unit, memory, display and operation module, and image processing and analysis software. The background server is used for remote background storage and management of image data collected in the field. The image collector of the mobile terminal adopts a camera with more than 5 million pixels, and the central processing unit adopts a quad-core CPU. Taking into account the image acquisition performance requirements

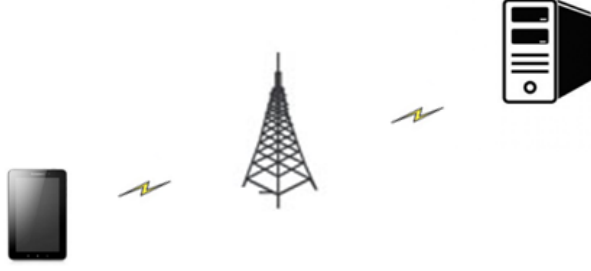
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**Figure 1: Deployment diagram of spider mite image acquisition system.**

of low power consumption, high stability, graphics processing, image transmission, and convenient portability, Intel chip board is selected as the hardware core. As for the main technical indicators: with main frequency up to 1.86GHz, supporting IPS touch screen, camera, it should support 3D acceleration GPU, have Bluetooth, WIFI, SIM functions, provide high-definition video HDMI interface. The display and operation module adopts an 8-inch touch screen. The memory includes built-in storage 32GB, and supports external expansion of TF card and U disk at the same time. To meet image processing and analysis software operation requirements, Win8 is used as the terminal's operating system.

### 3 SPIDER MITE IMAGE COMBINATION SEGMENTATION ALGORITHM

Due to the interference and influence of light, environment factors, the background of spider mite images collected in the field has certain complexity. It is difficult to achieve complete segmentation of spider mite images only by relying on a single color image segmentation method or gray threshold segmentation method. Therefore, image combination segmentation method combining K-means clustering analysis and Canny edge detection is adopted. That is, the K-means clustering analysis based on color space is used to filter out most leaf background and impurities of the collected image samples to obtain the spider mite grayscale images. Then, Canny operator edge detection algorithm is used to extract the closed contour curve of the spider mite image in the grayscale image. Finally, complete separation and extraction of the spider mite image is achieved through algorithms such as seed filling and morphological opening operations. The entire design and process of the combination segmentation method are shown in Fig 2

First, obtain the initial RGB color image of the plant leaf spider mite insect pest through the image acquisition device. Since the color of each pixel in the color image is determined by the three components of R, G, and B, and each component has 255 values, such a pixel point can have a color variation range of more than 16 million ( $255 \times 255 \times 255$ ). To reduce the calculation amount and segmentation difficulty in the subsequent image processing, the collected RGB color image is first converted into grayscale image. Due to the complexity of field-collected images, if we directly convert RGB images into grayscale images, it is not only impossible to remove background and impurities, but also most of the pest image

information will be lost. Considering that the pixel points of spider mites, leaves and impurities in the image display respective feature similarities in a certain color space, K-means clustering analysis as an unsupervised learning method can identify association rule from the feature data of the research object. Therefore, if we implement K-means clustering analysis based on this feature to convert color images into grayscale images, it is not only possible to remove most of the leaf background, impurities and other interference information around the spider mite image, but also the spider mite image information can be separated and enhanced.

When using K-means clustering analysis to process complex color images, if RGB values of pixel points are simply used as the feature vector to form the feature vector space, the effect is undesirable. Here, the RGB image samples are first converted to the color space Lab, and then the color, texture, and location characteristics of the pixels are extracted to form a feature vector.

Assume that a color image data set  $X = \{x_m | m = 1, 2, \dots, total\}$  is given, the samples in  $X$  are represented by  $d$  description attributes  $A_1, A_2, \dots, A_d$ , and the  $d$  description attributes are all continuous attributes. Data samples  $x_i = (x_{i1}, x_{i2}, \dots, x_{id})$ ,  $x_j = (x_{j1}, x_{j2}, \dots, x_{jd})$ . Where,  $x_{i1}, x_{i2}, \dots, x_{id}$  and  $x_{j1}, x_{j2}, \dots, x_{jd}$  are specific values of samples  $x_i$  and  $x_j$  with regard to  $d$  description attributes  $A_1, A_2, \dots, A_d$ .

The K-means clustering analysis algorithm is as follows:

- ① Determine an initial cluster center for each image data cluster, so that there are  $k$  initial cluster centers.
- ② Allocate the samples in the sample set to the nearest neighbor cluster according to the principle of minimum distance.
- ③ Calculate the sample mean in each cluster as the new cluster center.
- ④ Repeat steps ② and ③ until the cluster center no longer changes.
- ⑤ The clustering is over, and the spider mite grayscale image after removing most of the redundant leaf background is obtained.

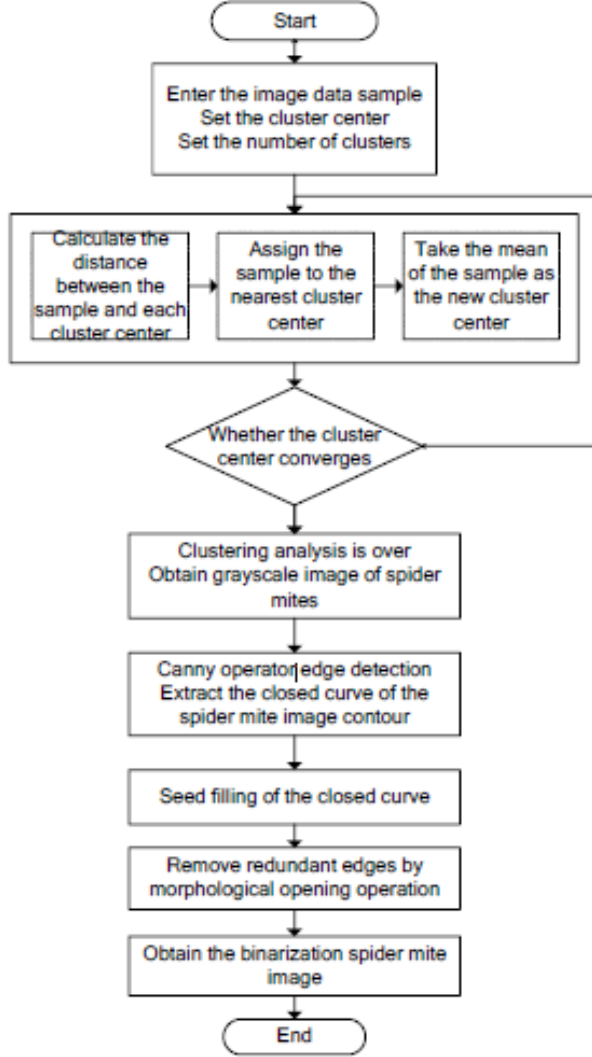
When calculating the distance between data samples  $x_i$  and  $x_j$ , the Euclidean distance algorithm is used as the sample similarity measure.

$$D(x_i, x_j) = \sqrt{\sum_{k=1}^d (x_{ik} - x_{jk})^2} \quad (1)$$

At the same time, error sum of squares criterion function is used to evaluate the clustering performance. Assume that  $X$  contains  $K$  cluster subsets  $X_1, X_2, \dots, X_k$ , the number of samples in each cluster subset is  $n_1, n_2, \dots, n_k$ , and the cluster centers of each cluster subset are  $m_1, m_2, \dots, m_k$ . Then the error sum of squares criterion function formula is:

$$E = \sum_{i=1}^k \sum_{p \in x_i} p - m_i^2 \quad (2)$$

In the grayscale image obtained by K-means clustering analysis, the pest image information is significantly enhanced. At this time, by using edge detection algorithm, a complete closed curve of the spider mite image edge can be obtained. Canny edge detection operator is a multi-scale spatial edge detection operator, its edge positioning accuracy is high, and there is certain ability to suppress noise. Therefore, after obtaining the grayscale image, Canny edge



**Figure 2: The flow chart of the combination segmentation algorithm for spider mite images.**

detection algorithm is used to extract the edge closed curve of spider mite image in the grayscale image.

The edge detection algorithm based on the Canny operator is as follows. The derivative operator is used to find the partial derivative ( $G_x$ ,  $G_y$ ) of the gray level in the grayscale image along two directions, and the gradient magnitude is calculated according to the following formula:

$$|G| = \sqrt{G_x^2 + G_y^2} \quad (3)$$

Gradient direction is calculated according to the following formula:

$$\theta = \text{Arctan} \left( \frac{G_y}{G_x} \right) \quad (4)$$

Considering that the edges in the grayscale image may point to different directions, the edges in the following four directions are

detected: horizontal direction, vertical direction,  $45^\circ$  angle from horizontal direction and  $135^\circ$  angle from horizontal direction. Then, for each pixel point in the grayscale image, compare it with the neighboring pixel points in the above direction, determine the edge direction generated by each pixel point to obtain the maximum grayscale of each pixel point and its neighboring pixel points. The maximum gray value is used as the maximum local gray value of each pixel point. Two thresholds are determined according to the maximum local grayscale of each pixel point: a high threshold and a low threshold. These two thresholds are used to determine clear edge pixel points. Since the important edges in the grayscale image are all continuous curves, it is possible to track the boundary part in the edge curve and avoid treating noise pixel points that do not form curves as edges. By picking a starting point, it is possible to track the fuzzy part of the edge curve until the starting point is restored. After completing the above process, the edge closed curve of the spider mite image can be extracted.

Finally, after the insect pest closed curve is obtained, the seed filling algorithm can be implemented on the area within the closed curve of the spider mite image. Here, four-connected domain technology is used for filling operation. As the spider mite image may have fuzzy edges and is subject to influence of neighboring impurities, the edges obtained after detection will have a small amount of redundant edges. Therefore, after filling the closed area with seeds, we use morphological opening operation to filter out redundant areas or edges in the grayscale image to obtain a complete spider mite binarization separation image, thereby achieving pest image segmentation.

## 4 SIMULATION TEST AND RESULT ANALYSIS

### 4.1 Combination Segmentation Method Test and Result Analysis

To verify the effect of the combination segmentation method, this method is used to perform segmentation experiments on the spider mite images in the collected image samples. The segmentation process and results are shown in Fig 3

Figure 3a is the original image of spider mite pests. By analyzing color RGB image of Figure 3a via K-means clustering algorithm, we obtain the grayscale separation image of Figure 3b. Where, the black dot image in the figure is the red spider mite image in the original image Figure 3a. After Canny edge detection and seed filling are performed on Figure 3b, the image in Figure 3c is obtained. By performing morphological opening operation on the image in Figure 3c, we obtain the binarization segmentation result in Figure 3d. The binarization image can be used for the subsequent calculation of insect number.

### 4.2 Analysis of Comparative Test Results

To verify the performance and segmentation effect of combination segmentation method, this method, the Lab space-based color segmentation method and the traditional otsu threshold segmentation method were used to perform image segmentation experiments on the leaf image samples of spider mite pests collected in the field to achieve segmentation of spider mite images. The segmentation effect of the three methods is shown in Figure 4

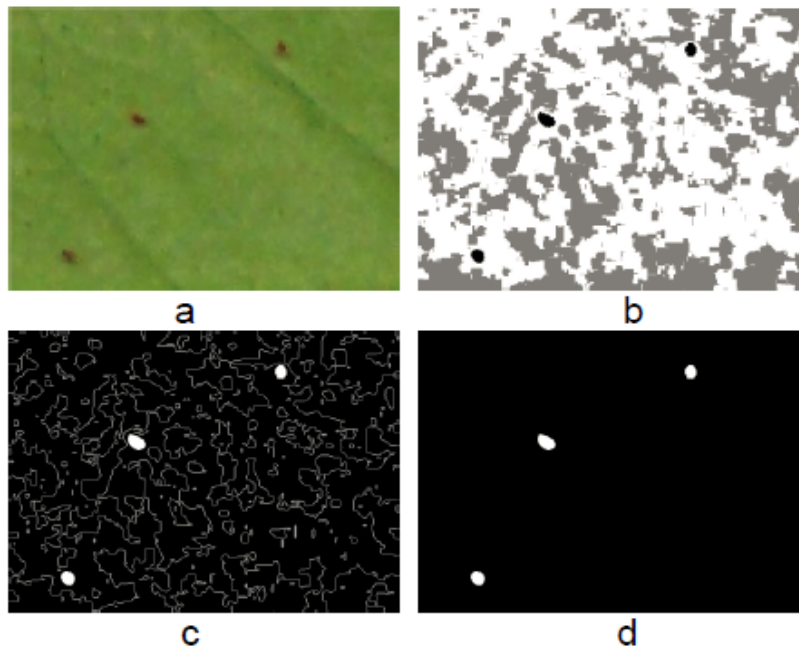


Figure 3: The segmentation effect of combination segmentation algorithm. a) Original image; b) K-means clustering analysis results; c) The effect of Canny edge detection and seed filling; d) The segmentation result after the morphological opening operation.

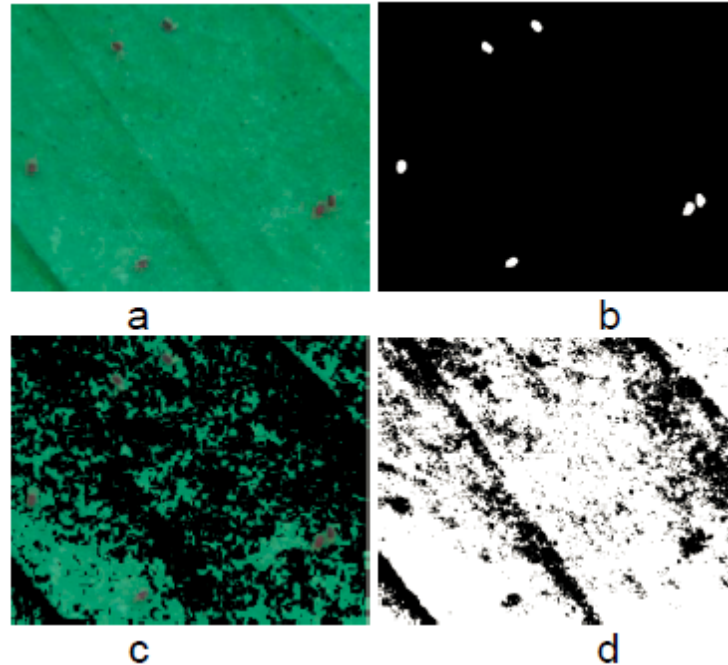


Figure 4: Comparison of the effect of the combination segmentation method and other segmentation methods. a) Original image b) Segmentation effect of combination segmentation method c) Lab space-based color segmentation effect d) Otsu threshold segmentation effect.

Figure 4a is the original image, Figure 4b is the result of the combination segmentation method, Figure 4c is the result of Lab space-based color segmentation method, and Figure 4d is the result of the otsu threshold segmentation method. Seen from the experimental results, it is difficult for traditional color segmentation method and threshold segmentation method to effectively distinguish between spider mite pest image and leaf background. Compared with traditional methods, the combination segmentation method combining clustering analysis and edge detection can not only effectively identify the characteristic information of spider mite pests, but also separate pest images from leaf background, which is conducive to subsequent image analysis and insect number calculation.

## 5 CONCLUSION

Based on the K-means clustering analysis of the color images of the plant leaf spider mites, spider mites image is separated from the leaf background via Canny edge detection algorithm. The separated images are subject to seed filling and morphological opening operations to effectively remove impurities and redundant edges generated by K-means clustering analysis and Canny edge detection, thereby achieving spider mite image segmentation. The test results and analysis show that based on the K-means clustering analysis and Canny edge detection algorithm, spider mite images on the leaves can be effectively segmented, which lays the foundation for subsequent pest analysis and automatic insect number counting. This method can be applied to the processing and analysis of spider mite stunts in the field, which is a new method for segmentation and recognition of field spider mite images. However, the actual field conditions of leaves damaged by spider mites are more complicated. Most spider mites have netting habit, and spider webs may interfere with image analysis. In addition, crops are often damaged by multiple plant diseases and insect pests at the same time, and leaf damage symptoms vary greatly, resulting in big changes in leaf

background. All these will affect the actual analysis results, which will be the directions for further subsequent research.

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